Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/08/2007
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Project Title:	The Use of Aspirin and Other NSAIDs to Ameliorate Muscle Atropy Due to Simulated Weightlessness		
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Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
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Space Biology Element:	None		
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Zip Code:	02215	**Congressional District:**	8
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed end date from 9/30/2006 to 10/1/06 to match fiscal year (1/09)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kandarian, Susan (MENTOR/Boston University)		
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	POSTDOCTORAL FELLOWSHIP The overall objective of the present application is to test the ability of aspirin and other non-steroidal anti-inflammatory drugs (NSAID) to inhibit skeletal muscle atrophy, weakness, and changes in gene expression due to simulated weightlessness in rats using the hind limb suspension model. The specific aims for this proposal are to: 1) determine if high dose aspirin treatment inhibits: soleus and plantaris fiber atrophy, deficits in maximum force, and the activation of NF-kappa B seen after 7 days of hindlimb unloading. The inhibition of cyclooxygenase activity by NSAID treatment will also be assessed in muscles; 2) determine if curcumin or naproxen treatment inhibits: soleus and plantaris fiber atrophy, deficits in maximum force, and NF-kappa B or cyclooxygenase activity due to 7 days of unloading; 3) determine if the expression of genes known to be upregulated with 7 days of unloading are “normalized” by treatment		

Task Description:	with any of the NSAIDs. To determine if curcumin inhibits NF-kappa B activity, C2C12 mouse myotubes were treated with 10 ngrams/ml TNF-alpha (a potent activator of NF-kappa B) for 3 hours either in the presence or absence of 200 mM curcumin. While TNF-alpha caused a 9-fold increase in NF-kappa B activity, curcumin completely abolished this increase. These results suggest that curcumin may be an effective countermeasure. However, in hind limb unloaded animals injection of curcumin did not attenuate the increase in NF-kapp B activity, nor did it attenuate the unloading-induced muscle atrophy. While disappointing, this is in agreement with a very recent study showing dietary curcumin does not attenuate unloading-induced NF-kappa B activity in skeletal muscle or unloading-induced skeletal muscle dysfunction or atrophy (Farid et al. Effects of dietary curcumin or N-acetylcysteine on NF-?B activity and contractile performance in ambulatory and unloaded murine soleus. Nutr Metab (Lond). 2005; 2: 20). The ineffectiveness of curcumin in vivo may be due to the low bioavailability. To determine if aspirin inhibits NF-kappa B activity, C2C12 mouse myotubes were treated with 10 ngrams/ml TNF-alpha for 3 hours either in the presence or absence of 20mM aspirin. TNF-alpha caused an 11-fold increase in NF-kappa B activity and aspirin abolished this increase. In unloaded animals, injection of aspirin did not attenuate the increase in NF-kappa B activity, but it did attenuate the unloading-induced muscle atrophy. Therefore, this dose of aspirin may be working by the inhibition of other molecules, which we have yet to test. To determine if naproxen inhibits muscle atrophy and NF-kappa B activity we injected 10 mg/kg naproxen twice daily, beginning 1 day prior to hind limb unloading and for the duration of the unloading period. The results showed that this dose of naproxen did not inhibit NF-kappa B activity or muscle atrophy, due to hind limb unloading. To gain further insight into other components of the NF-kappa B pathway that are necessary for unloading-induced atrophy a fourth and fifth aim were added using genetic approaches to focus on the I kappa B alpha protein (which is downstream of I kappa B kinase) and the c-rel gene. By determining the involvement of particular proteins and genes in the atrophy process, specific pharmacological countermeasures may be tested in subsequent studies. Aim 4: To determine if I kappa B alpha is required for unloading-induced NF-?B activity, muscle atrophy, and upregulation of atrophy-related genes. For this aim an I kappa B alpha dominant negative plasmid (known as a “superrepressor”) was injected into the soleus muscle of rats. This superrepressor is resistant to phosphorylation by I kappa B kinase and therefore resistant to ubiquitination and subsequent degradation. NF-kappa B activity was increased 5-fold with hind limb unloading, but completely abolished in the unloaded muscles injected with the superrepressor. Furthermore, soleus muscle fiber cross sectional area was decreased by 40% following 7 days unloading but was attenuated by 40% in the unloaded muscles injected with the superrepressor. In addition the increase in gene expression of atrogin-1/MAFbx, Cathepsin L, Nedd4, IEX-1, 4E-BP1 and FOXO3a with unloading was significantly attenuated in the unloaded muscles injected with the superrepressor. Aim 5: To determine if c-rel is required for unloading-induced NF-kappa B activity and muscle atrophy. For this aim we unloaded wild type (WT) and c-rel knockout (-/-) mice. NF-kappa B activity was increased 8-fold, with unloading, in the soleus muscle of WT mice, and remained equally elevated with unloading in the c-rel -/- mice. Moreover, muscle atrophy was the same in WT unloaded and c-rel -/- unloaded. This eliminates c-rel as a contributor to the atrophy process and, therefore, as a therapeutic target. However, one potential target emerged from our previously published data, using Nfkb1 knockout mice (which encodes for the NF-kappaB family member p50). In this study we demonstrated that p50 is necessary for unloading-induced muscle atrophy and NF-kappaB activation. Based on this data we chose to add one further countermeasure - andrographolide - that has been shown to inhibit p50-DNA binding. To determine if andrographolide inhibits NF-kappa B activity in our hands, we treated C2C12 mouse myotube with 10 ngrams/ml TNF-alpha for 3 hours either in the presence or absence of 50µM andrographolide. TNF-alpha caused an ~17 fold increase in NF-kappa B activity, which was abolished with andrographolide. Since andrographolide is only soluble in DMSO, we chose to supplement the rat chow with 20 mg/kg/day andrographolide, instead of injecting the countermeasure. Andrographolide supplementation inhibited NF-kappa B activation by 28% and muscle atrophy by 27%, following 7 days of unloading, suggesting that andrographolide could be a suitable countermeasure. However, further investigation is clearly necessary.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	By identifying specific proteins and genes involved in skeletal muscle atrophy and testing nutritional and/or pharmacological countermeasures to them we will be in a much better position to combat the deleterious changes in muscle function due to any type of disuse.
Task Progress:	To determine if naproxen inhibits muscle atrophy and NF-kappa B activity we injected 10 mg/kg naproxen twice daily, beginning 1 day prior to hind limb unloading and for the duration of the unloading period. The results showed that this dose of naproxen did not inhibit NF-kappa B activity or muscle atrophy, due to hind limb unloading. However, one potential target emerged from our previously published data, using Nfkb1 knockout mice (which encodes for the NF-kappaB family member p50). In this study we demonstrated that p50 is necessary for unloading-induced muscle atrophy and NF-kappaB activation. Based on this data we chose to add one further countermeasure - andrographolide - that has been shown to inhibit p50-DNA binding. To determine if andrographolide inhibits NF-kappa B activity in our hands, we treated C2C12 mouse myotube with 10 ngrams/ml TNF-alpha for 3 hours either in the presence or absence of 50µM andrographolide. TNF-alpha caused an ~17 fold increase in NF-kappa B activity, which was abolished with andrographolide. Since andrographolide is only soluble in DMSO, we chose to supplement the rat chow with 20 mg/kg/day andrographolide, instead of injecting the countermeasure. Andrographolide supplementation inhibited NF-kappa B activation by 28% and muscle atrophy by 27%, following 7 days of unloading, suggesting that andrographolide could be a suitable countermeasure. However, further investigation is clearly necessary.
Bibliography Type:	Description: (Last Updated: 05/07/2007)

Articles in Peer-reviewed Journals	Judge AR, Koncarevic A, Hunter RB, Liou HC, Jackman RW, Kandarian SC. "Role for IkappaBalpha, but not c-Rel, in skeletal muscle atrophy." Am J Physiol Cell Physiol. 2007 Jan;292(1):C372-82. PMID: 16928772 , Jan-2007